

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : Retail フリーザー クリーナー
RETAIL FREEZER/COOLER CLEANER

Other means of identification : Not applicable

Recommended use : Cleaning product

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : Product is sold ready to use.

Company : Kay Chemical Company c/o Ecolab G.K.
1-31-18, Takadanobaba, Shinjuku-ku Tokyo
169-8631 Japan
0120-103-932 (Toll Free)

Emergency telephone : 0120-756-005 (Toll Free)

Issuing date : 2017.07.31

SECTION 2. HAZARDS IDENTIFICATION
GHS Classification

Flammable liquids : Category 3
Serious eye damage/eye irritation : Category 2B

GHS label elements

Hazard pictograms :



Signal Word : Warning

Hazard Statements : Flammable liquid and vapor.
Causes eye irritation.

Precautionary Statements : **Prevention:**
Keep away from heat/sparks/open flames/hot surfaces. No smoking.
Keep container tightly closed. Ground/bond container and receiving equipment. Use explosion-proof electrical/ ventilating/ lighting/ equipment. Use only non-sparking tools. Take precautionary measures against static discharge. Wash skin thoroughly after handling.
Response:
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/ attention. In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.
Storage:
Store in a well-ventilated place. Keep cool.

SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

Disposal:

Dispose of contents/ container to an approved waste disposal plant.

Other hazards : None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
polyol	Proprietary Ingredient	30 - 60
Isopropanol	67-63-0	5 - 10
monoethanolamine	141-43-5	1 - 10
Fattyalcohol ethoxylates < 12EO	Proprietary Ingredient	1 - 5

SECTION 4. FIRST AID MEASURES

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

Protection of first-aiders : No special precautions are necessary for first aid responders.

Notes to physician : Treat symptomatically.

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : High volume water jet

Specific hazards during fire fighting : Fire Hazard
Keep away from heat and sources of ignition.
Flash back possible over considerable distance.
Beware of vapors accumulating to form explosive concentrations.
Vapors can accumulate in low areas.

Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides
Nitrogen oxides (NOx)

Special protective equipment for fire-fighters : Use personal protective equipment.

Specific extinguishing methods : Use water spray to cool unopened containers. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

SECTION 6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Remove all sources of ignition. Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Eliminate all ignition sources if safe to do so. Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

- Advice on safe handling : Keep away from fire, sparks and heated surfaces. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapors). Wash hands thoroughly after handling.
- Conditions for safe storage : Keep away from heat and sources of ignition. Keep in a cool, well-ventilated place. Keep away from oxidizing agents. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.
- Storage temperature : 0 °C to 35 °C

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Threshold limit value and permissible exposure limits for each component in the work environment

Ingredients	CAS-No.	Form of exposure	Permissible concentration	Basis
glycerin	56-81-5	TWA	10 mg/m3	ACGIH
		TWA (respirable fraction)	5 mg/m3	OSHA Z1
Isopropanol	67-63-0	ACL	200 ppm	JP OEL ISHL
		OEL-C	400 ppm 980 mg/m3	JP OEL JSOH
Isopropanol	67-63-0	TWA	200 ppm	ACGIH
		STEL	400 ppm	ACGIH
		TWA	400 ppm 980 mg/m3	NIOSH REL
		STEL	500 ppm 1,225 mg/m3	NIOSH REL
		TWA	400 ppm 980 mg/m3	OSHA Z1
monoethanolamine	141-43-5	OEL-M	3 ppm 7.5 mg/m3	JP OEL JSOH
monoethanolamine	141-43-5	TWA	3 ppm	ACGIH
		STEL	6 ppm	ACGIH

SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

		TWA	3 ppm 8 mg/m ³	NIOSH REL
		STEL	6 ppm 15 mg/m ³	NIOSH REL
		TWA	3 ppm 6 mg/m ³	OSHA Z1

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: liquid
Color	: cloudy, light green
Odor	: alcohol-like
pH	: 10.6 - 11.8, (100 %)
Flash point	: 41 °C closed cup, Sustains combustion
Odor Threshold	: No data available
Melting point/freezing point	: No data available
Initial boiling point and boiling range	: No data available
Evaporation rate	: No data available
Flammability (solid, gas)	: No data available
Upper explosion limit	: No data available
Lower explosion limit	: No data available
Vapor pressure	: No data available
Relative vapor density	: No data available
Relative density	: 1.065 - 1.085
Water solubility	: soluble
Solubility in other solvents	: No data available
Partition coefficient: n-octanol/water	: No data available
Autoignition temperature	: No data available
Thermal decomposition	: No data available
Viscosity, kinematic	: No data available

SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

Explosive properties	: No data available
Oxidizing properties	: No data available
Molecular weight	: No data available
VOC	: No data available

SECTION 10. STABILITY AND REACTIVITY

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: Heat, flames and sparks.
Incompatible materials	: Acids
Hazardous decomposition products	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NOx)

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Eyes	: Causes eye irritation.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact	: Redness, Irritation
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Toxicity

Product

Acute oral toxicity	: Acute toxicity estimate : > 2,000 mg/kg
Acute inhalation toxicity	: 4 h Acute toxicity estimate : > 5 mg/l
Acute dermal toxicity	: Acute toxicity estimate : > 2,000 mg/kg
Skin corrosion/irritation	: Mild skin irritation

SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

Serious eye damage/eye irritation	: Mild eye irritation Method: OECD Test Guideline 437 Test substance: Product
Respiratory or skin sensitization	: No data available
Carcinogenicity	: No data available
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: No data available
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Environmental Effects : This product has no known ecotoxicological effects.

Product

Toxicity to fish	: No data available
Toxicity to daphnia and other aquatic invertebrates	: No data available
Toxicity to algae	: No data available

Ingredients

Toxicity to fish	: polyol 96 h LC50 Fish: 855 mg/l Isopropanol 96 h LC50 Pimephales promelas: 9,640 mg/l Fattyalcohol ethoxylates < 12EO 96 h LC50 Fish: 3.2 mg/l
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Ingredients

Toxicity to daphnia and other aquatic invertebrates	: Isopropanol LC50 Daphnia magna (Water flea): > 10,000 mg/l monoethanolamine 48 h EC50 Daphnia: 65 mg/l
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Persistence and degradability

No data available

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

- Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

SECTION 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

- UN number : 1993
- Description of the goods : FLAMMABLE LIQUID, N.O.S.
(Isopropanol)
- Class : 3
- Packing group : III
- Environmentally hazardous : no

Sea transport (IMDG/IMO)

- UN number : 1993
- Description of the goods : FLAMMABLE LIQUID, N.O.S.
(Isopropanol)
- Class : 3
- Packing group : III
- Marine pollutant : no

SECTION 15. REGULATORY INFORMATION

National regulatory information

Fire Service Law

Group 4, Type 2 petroleums, Hazardous rank III, Water soluble liquid, Keep away from fire, (2000 liter)

Industrial Safety and Health Law

Prohibited Substances

Not applicable

Harmful Substances Subject to Obtain the Permission of Manufacturing SDS Table 3-1

Not applicable

Publication of Technical Guidelines Article 28 (3)

Not applicable

Mutagenic existing chemical substances

Not applicable

Mutagenic new chemical substances

SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

Not applicable

Hazardous Substances Requiring Notification SDS Table 9

Article 57-2 (Enforcement Order Table 9)

Chemical name	Concentration (%)
Isopropanol	5 - 10
monoethanolamine	1 - 5

Hazardous Substances Subject to Labeling Requirements

Article 57 (Enforcement Order Article 18)

Chemical name
Isopropanol
monoethanolamine

Ordinance on Prevention of Specified Chemical Substances Hazards

Not applicable

Ordinance on Prevention of Lead Poisoning

Not applicable

Ordinance on Prevention of Tetraalkyl Lead Poisoning

Not applicable

Ordinance on Prevention of Organic Solvent Poisoning

Organic Solvents Class 2

Enforcement Order of the Industrial Safety and Health Law: Dangerous Substances

Inflammable Substance

Poisonous and Deleterious Substances Control Law

Not applicable

Law concerning Pollutant Release and Transfer Register (PRTR Law)

Class I Designated Chemical Substances

Chemical name	Concentration (%)
monoethanolamine	3.0

Water Pollution Control Law

Not applicable

The ingredients of this product are reported in the following inventories:

United States TSCA Inventory :

On TSCA Inventory

Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL

Australia Inventory of Chemical Substances (AICS) :

SAFETY DATA SHEET

RETAIL FREEZER/COOLER CLEANER

On the inventory, or in compliance with the inventory

New Zealand. Inventory of Chemical Substances :

On the inventory, or in compliance with the inventory

Japan. ENCS - Existing and New Chemical Substances Inventory :

On the inventory, or in compliance with the inventory

Korea. Korean Existing Chemicals Inventory (KECI) :

On the inventory, or in compliance with the inventory

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :

On the inventory, or in compliance with the inventory

China. Inventory of Existing Chemical Substances in China (IECSC) :

On the inventory, or in compliance with the inventory

Taiwan Chemical Substance Inventory :

On the inventory, or in compliance with the inventory

SECTION 16. OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet
Globally Harmonized System of Classification and Labeling of Chemicals (GHS)
IARC: (International Agency for Research on Cancer)
US. National Toxicology Program (NTP) Report on Carcinogens
ECHA List of Publishable Substances Registered
EU HPVCs (High Production Volume Chemicals)

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Version	: 1.3
Prepared by	: Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.